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*Series and sums $k=0$ to n k & $= \frac{n(n+1)}{2}$, & $k=0$ to n k^2 & $= \frac{n(n+1)(2n+1)}{6}$, $\sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$.

*Integrals $\int_a^b x^n dx = \frac{b^{n+1}-a^{n+1}}{n+1}$, $\int_a^b \frac{1}{x} dx = \ln b - \ln a$,
 $\int_a^b e^{cx} dx = \frac{e^{cx}}{c}$,
 $\int_a^b \ln x dx = x \ln x - x$,

Integration by parts: $\int u dv = uv - \int v du$.

*Piecewise definitions

$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$ $\operatorname{sgn}(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$

*Limits and derivatives

$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x)$, $\frac{d}{dx} (f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$.

Chain rule: $\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$. Taylor's theorem with remainder:

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1}$$
 for some ξ between a and x .

*Matrices For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$\det A = ad - bc$, $A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$, $\operatorname{tr} A = a + d$.

The 3×3 identity and a general matrix:

$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $M = \begin{pmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{pmatrix}$

Eigenvalues satisfy $\det(A - \lambda I) = 0$; for symmetric $A \in \mathbb{R}^{n \times n}$ all $\lambda_i \in \mathbb{R}$ and $A = Q \Lambda Q^T$ with Q orthogonal.

*Inequalities $|a+b| \leq |a| + |b|$, $|a-b| \leq |a| + |b|$, $|a| \leq |b|$ implies $|a-b| \leq |b|$. Triangle inequality: $|x+y| \leq |x| + |y|$ for all $x, y \in \mathbb{R}^n$.

*Greek letters and operators $\alpha, \beta, \gamma, \delta, \epsilon, \vartheta, \lambda, \mu, \pi, \sigma, \varphi, \omega, \Gamma, \Delta, \Theta$.
 Operators and relations: $\partial, \nabla, \infty, \pm, \times, \circ, \cdot, \leq, \geq, \neq, \approx, \equiv, \subseteq, \sup, \cap, \emptyset$.